FAIRCHILD SEMICONDUCTOR®

HGTD7N60A4S, HGT1S7N60A4S, HGTG7N60A4, HGTP7N60A4

Data Sheet **December 2001**

600V, SMPS Series N-Channel IGBT

The HGTD7N60A4S, HGT1S7N60A4S, HGTG7N60A4 and HGTP7N60A4 are MOS gated high voltage switching devices combining the best features of MOSFETs and bipolar transistors. These devices have the high input impedance of a MOSFET and the low on-state conduction loss of a bipolar transistor. The much lower on-state voltage drop varies only moderately between $25^{o}C$ and $150^{o}C$.

This IGBT is ideal for many high voltage switching applications operating at high frequencies where low conduction losses are essential. This device has been optimized for high frequency switch mode power supplies.

Formerly Developmental Type TA49331.

Ordering Information

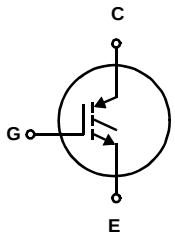
PART NUMBER	PACKAGE	BRAND
HGTD7N60A4S	TO-252AA	7N60A4
HGT1S7N60A4S	TO-263AB	7N60A4
HGTG7N60A4	TO-247	7N60A4
HGTP7N60A4	TO-220AB	7N60A4

NOTE: When ordering, use the entire part number. Add the suffix 9A to obtain the TO-252AA and TO-263AB variant in tape and reel, e.g., HGTD7N60A4S9A.

Features

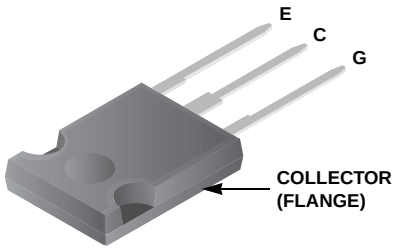
- >100kHz Operation at 390V, 7A
- 200kHz Operation at 390V, 5A
- 600V Switching SOA Capability
- Typical Fall Time. . . . . 75ns at $T_J = 125^{o}C$
- Low Conduction Loss
- *Temperature Compensating* SABER™ Model www.Fairchild.com

Symbol

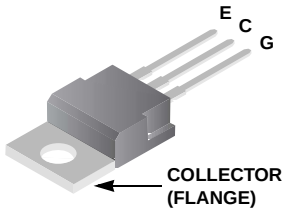


Packaging

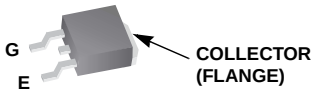
JEDEC STYLE TO-247



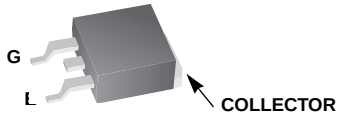
JEDEC TO-220AB



JEDEC TO-252AA



JEDEC TO-263AB



Fairchild CORPORATION IGBT PRODUCT IS COVERED BY ONE OR MORE OF THE FOLLOWING U.S. PATENTS

4,364,073	4,417,385	4,430,792	4,443,931	4,466,176	4,516,143	4,532,534	4,587,713
4,598,461	4,605,948	4,620,211	4,631,564	4,639,754	4,639,762	4,641,162	4,644,637
4,682,195	4,684,413	4,694,313	4,717,679	4,743,952	4,783,690	4,794,432	4,801,986
4,803,533	4,809,045	4,809,047	4,810,665	4,823,176	4,837,606	4,860,080	4,883,767
4,888,627	4,890,143	4,901,127	4,904,609	4,933,740	4,963,951	4,969,027	

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Absolute Maximum Ratings $T_C = 25^oC$, Unless Otherwise Specified

		ALL TYPES	UNITS
Collector to Emitter Voltage	BV_{CES}	600	V
Collector Current Continuous			
At $T_C = 25^oC$	I_{C25}	34	A
At $T_C = 110^oC$	I_{C110}	14	A
Collector Current Pulsed (Note 1)	I_{CM}	56	A
Gate to Emitter Voltage Continuous	V_{GES}	±20	V
Gate to Emitter Voltage Pulsed	V_{GEM}	±30	V
Switching Safe Operating Area at $T_J = 150^oC$, Figure 2	SSOA	35A at 600V	
Single Pulse Avalanche Energy at $T_C = 25^oC$	E_{AS}	25mJ at 7A	
Power Dissipation Total at $T_C = 25^oC$	P_D	125	W
Power Dissipation Derating $T_C > 25^oC$		1.0	$W/^oC$
Operating and Storage Junction Temperature Range	T_J, T_{STG}	-55 to 150	^oC
Maximum Lead Temperature for Soldering			
Leads at 0.063in (1.6mm) from Case for 10s	T_L	300	^oC
Package Body for 10s, See Tech Brief 334	T_{PKG}	260	^oC

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

NOTE:

1. Pulse width limited by maximum junction temperature.

Electrical Specifications $T_J = 25^oC$, Unless Otherwise Specified

PARAMETER	SYMBOL	TEST CONDITIONS		MIN	TYP	MAX	UNITS
Collector to Emitter Breakdown Voltage	BV_{CES}	$I_C = 250\mu A$, $V_{GE} = 0V$		600	-	-	V
Emitter to Collector Breakdown Voltage	BV_{ECS}	$I_C = 10mA$, $V_{GE} = 0V$		20	-	-	V
Collector to Emitter Leakage Current	I_{CES}	$V_{CE} = 600V$	$T_J = 25^oC$	-	-	250	μA
			$T_J = 125^oC$	-	-	2	mA
Collector to Emitter Saturation Voltage	$V_{CE(SAT)}$	$I_C = 7A$, $V_{GE} = 15V$	$T_J = 25^oC$	-	1.9	2.7	V
			$T_J = 125^oC$	-	1.6	2.2	V
Gate to Emitter Threshold Voltage	$V_{GE(TH)}$	$I_C = 250\mu A$, $V_{CE} = 600V$		4.5	5.9	7.0	V
Gate to Emitter Leakage Current	I_{GES}	$V_{GE} = \pm 20V$		-	-	±250	nA
Switching SOA	SSOA	$T_J = 150^oC$, $R_G = 25\Omega$, $V_{GE} = 15V$, $L = 100\mu H$, $V_{CE} = 600V$		35	-	-	A
Pulsed Avalanche Energy	E_{AS}	$I_{CE} = 7A$, $L = 500\mu H$		25	-	-	mJ
Gate to Emitter Plateau Voltage	V_{GEP}	$I_C = 7A$, $V_{CE} = 300V$		-	9.0	-	V
On-State Gate Charge	$Q_{g(ON)}$	$I_C = 7A$, $V_{CE} = 300V$	$V_{GE} = 15V$	-	37	45	nC
			$V_{GE} = 20V$	-	48	60	nC
Current Turn-On Delay Time	$t_{d(ON)I}$	IGBT and Diode at $T_J = 25^oC$, $I_{CE} = 7A$, $V_{CE} = 390V$, $V_{GE} = 15V$, $R_G = 25\Omega$, $L = 1mH$, Test Circuit (Figure 20)		-	11	-	ns
Current Rise Time	t_{rI}			-	11	-	ns
Current Turn-Off Delay Time	$t_{d(OFF)I}$			-	100	-	ns
Current Fall Time	t_{fI}			-	45	-	ns
Turn-On Energy (Note 2)	E_{ON1}			-	55	-	μJ
Turn-On Energy (Note 2)	E_{ON2}			-	120	150	μJ
Turn-Off Energy (Note 3)	E_{OFF}			-	60	75	μJ

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Electrical Specifications $T_J = 25^oC$, Unless Otherwise Specified **(Continued)**

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
Current Turn-On Delay Time	$t_{d(ON)I}$	IGBT and Diode at $T_J = 125^oC$	-	10	-	ns
Current Rise Time	t_{rI}	$I_{CE} = 7A$	-	7	-	ns
Current Turn-Off Delay Time	$t_{d(OFF)I}$	$V_{CE} = 390V$	-	130	150	ns
Current Fall Time	t_{fI}	$V_{GE} = 15V$	-	75	85	ns
Turn-On Energy (Note 2)	E_{ON1}	$R_G = 25\Omega$	-	50	-	µJ
Turn-On Energy (Note 2)	E_{ON2}	$L = 1mH$	-	200	215	µJ
Turn-Off Energy (Note 3)	E_{OFF}	Test Circuit (Figure 20)	-	125	170	µJ
Thermal Resistance Junction To Case	$R_{\theta JC}$		-	-	1.0	$^oC/W$

NOTES:

2. Values for two Turn-On loss conditions are shown for the convenience of the circuit designer. E_{ON1} is the turn-on loss of the IGBT only. E_{ON2} is the turn-on loss when a typical diode is used in the test circuit and the diode is at the same T_J as the IGBT. The diode type is specified in Figure 20.
3. Turn-Off Energy Loss (E_{OFF}) is defined as the integral of the instantaneous power loss starting at the trailing edge of the input pulse and ending at the point where the collector current equals zero ($I_{CE} = 0A$). All devices were tested per JEDEC Standard No. 24-1 Method for Measurement of Power Device Turn-Off Switching Loss. This test method produces the true total Turn-Off Energy Loss.

Typical Performance Curves Unless Otherwise Specified

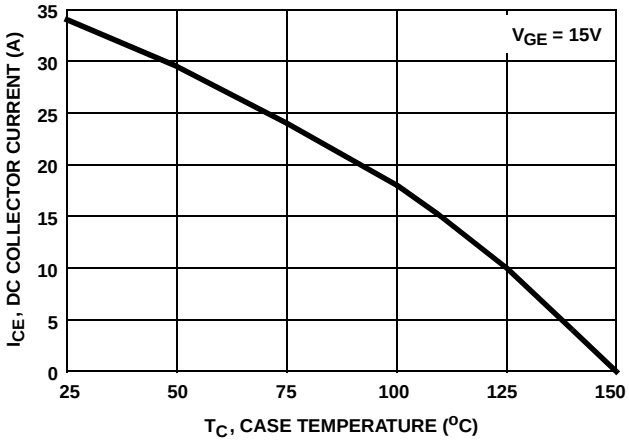


FIGURE 1. DC COLLECTOR CURRENT vs CASE TEMPERATURE

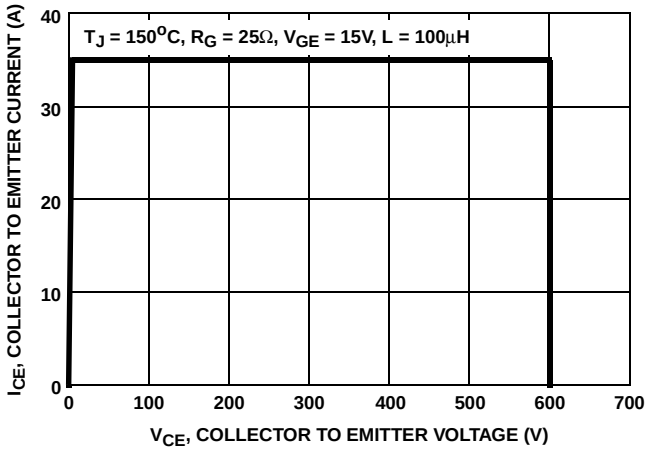


FIGURE 2. MINIMUM SWITCHING SAFE OPERATING AREA

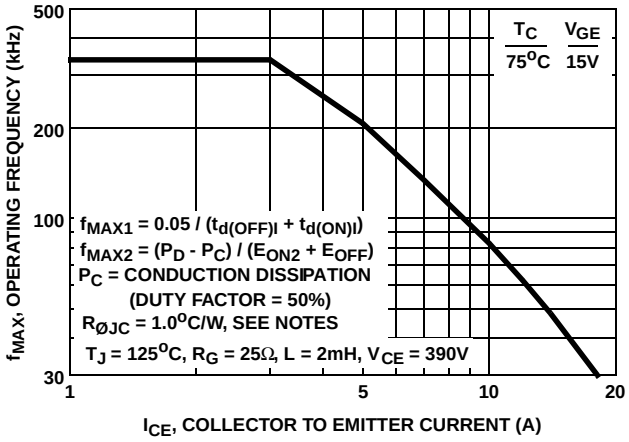


FIGURE 3. OPERATING FREQUENCY vs COLLECTOR TO EMITTER CURRENT

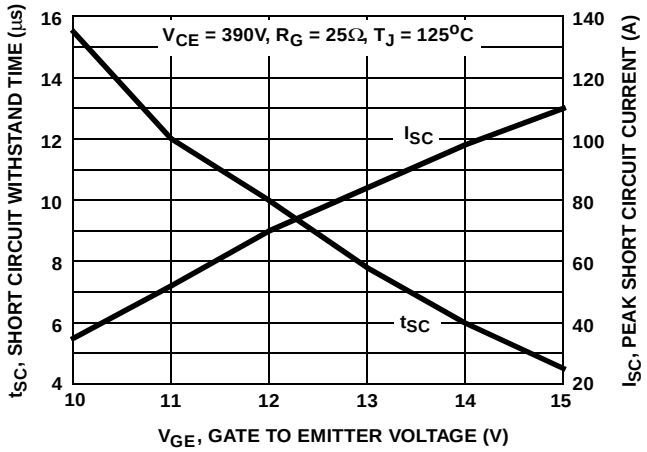


FIGURE 4. SHORT CIRCUIT WITHSTAND TIME

Typical Performance Curves Unless Otherwise Specified (Continued)

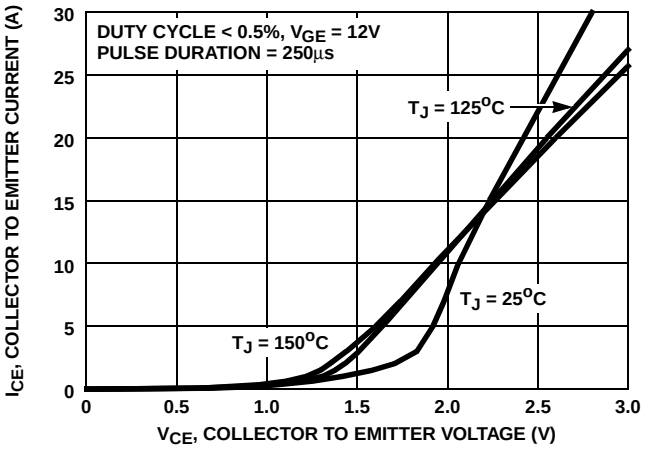


FIGURE 5. COLLECTOR TO EMITTER ON-STATE VOLTAGE

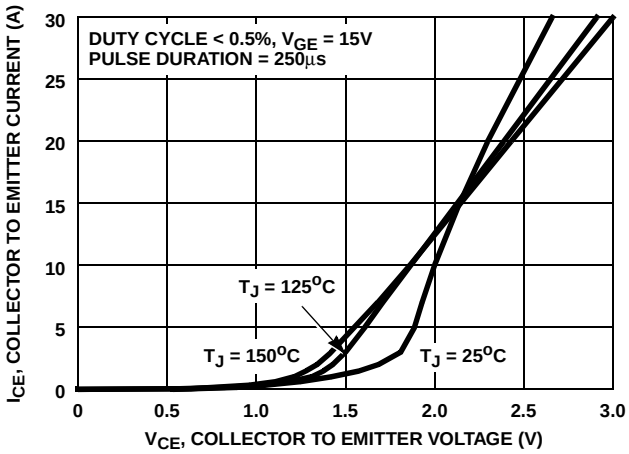


FIGURE 6. COLLECTOR TO EMITTER ON-STATE VOLTAGE

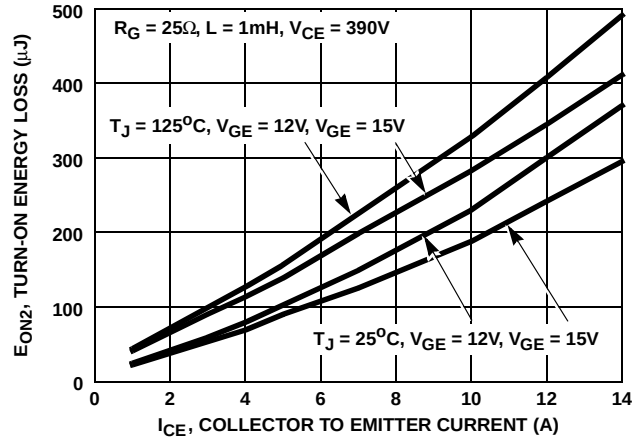


FIGURE 7. TURN-ON ENERGY LOSS vs COLLECTOR TO EMITTER CURRENT

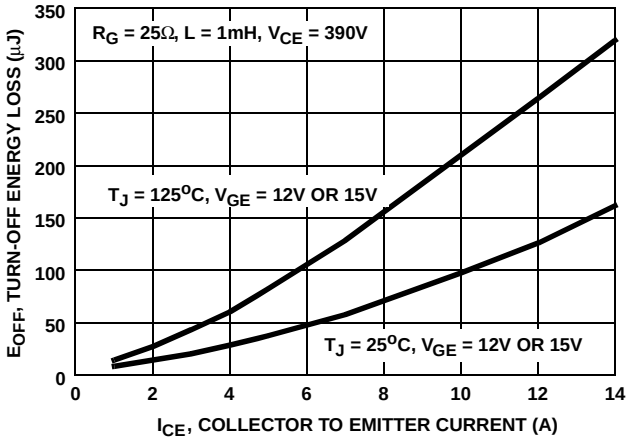


FIGURE 8. TURN-OFF ENERGY LOSS vs COLLECTOR TO EMITTER CURRENT

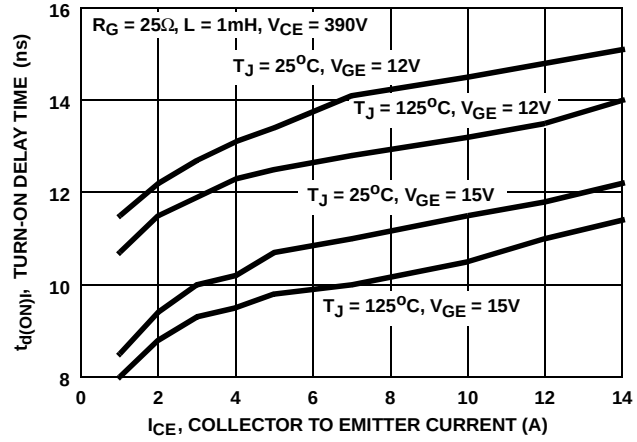


FIGURE 9. TURN-ON DELAY TIME vs COLLECTOR TO EMITTER CURRENT

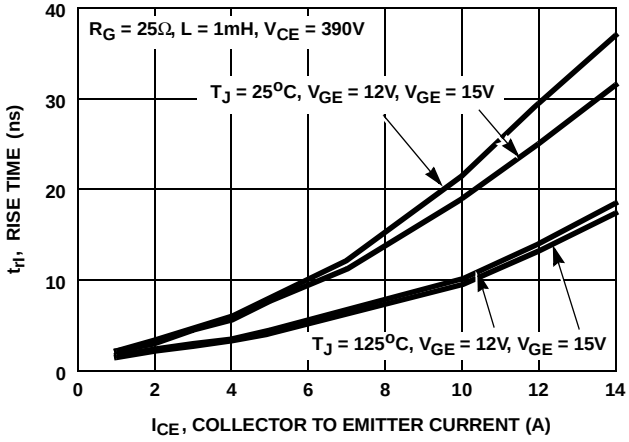


FIGURE 10. TURN-ON RISE TIME vs COLLECTOR TO EMITTER CURRENT

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Typical Performance Curves Unless Otherwise Specified (Continued)

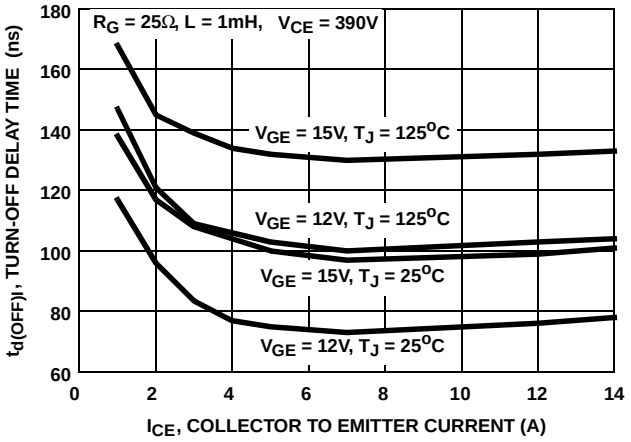


FIGURE 11. TURN-OFF DELAY TIME vs COLLECTOR TO EMITTER CURRENT

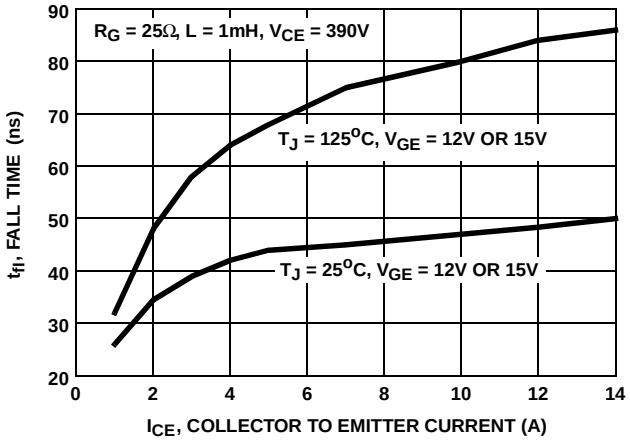


FIGURE 12. FALL TIME vs COLLECTOR TO EMITTER CURRENT

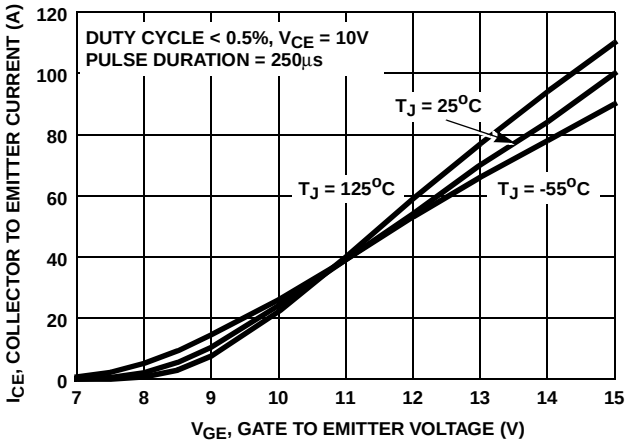


FIGURE 13. TRANSFER CHARACTERISTIC

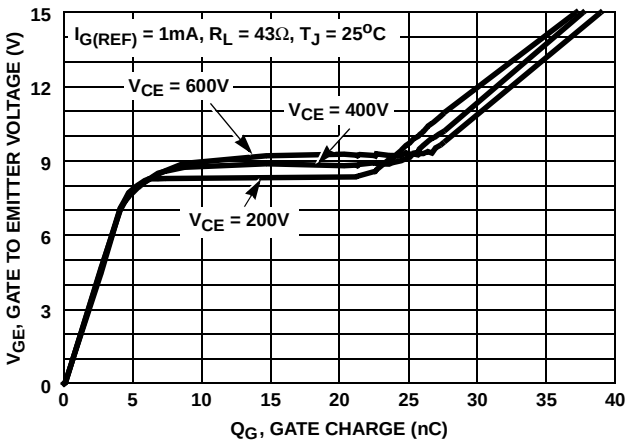


FIGURE 14. GATE CHARGE WAVEFORMS

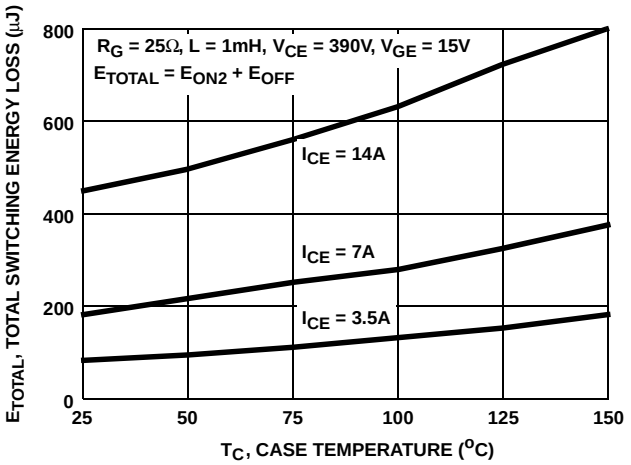


FIGURE 15. TOTAL SWITCHING LOSS vs CASE TEMPERATURE

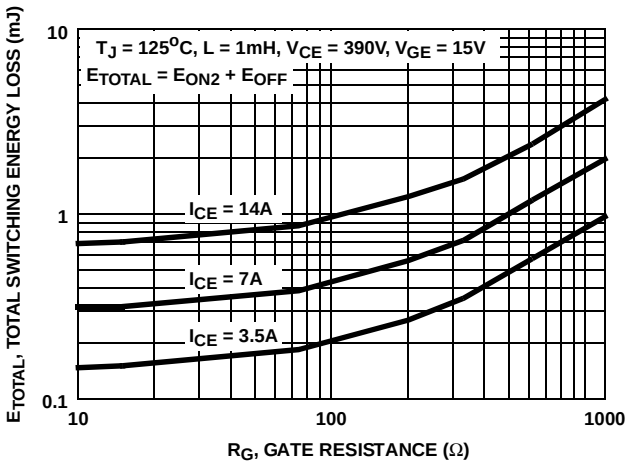


FIGURE 16. TOTAL SWITCHING LOSS vs GATE RESISTANCE

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Typical Performance Curves Unless Otherwise Specified (Continued)

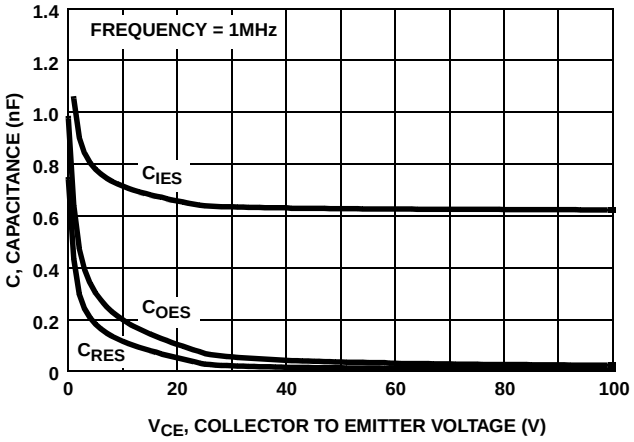


FIGURE 17. CAPACITANCE vs COLLECTOR TO EMITTER VOLTAGE

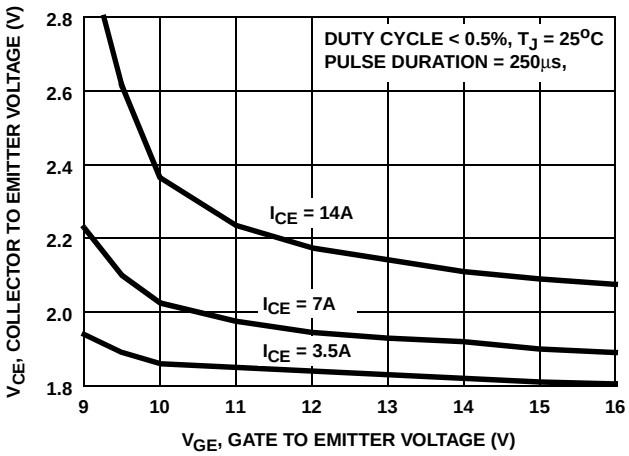


FIGURE 18. COLLECTOR TO EMITTER ON-STATE VOLTAGE vs GATE TO EMITTER VOLTAGE

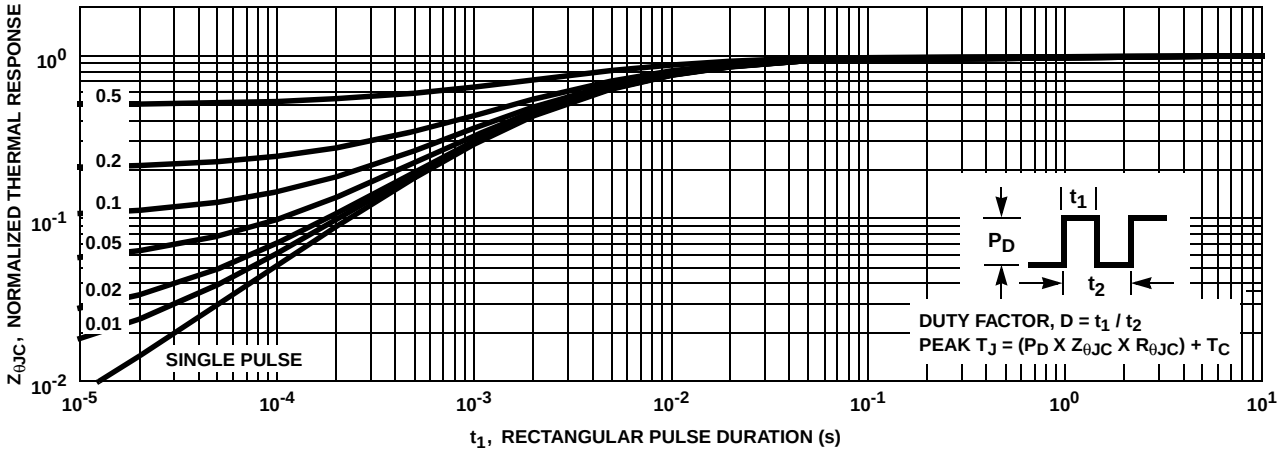


FIGURE 19. IGBT NORMALIZED TRANSIENT THERMAL RESPONSE, JUNCTION TO CASE

Test Circuit and Waveforms

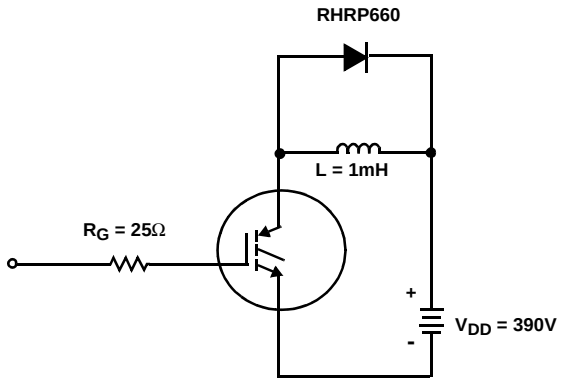


FIGURE 20. INDUCTIVE SWITCHING TEST CIRCUIT

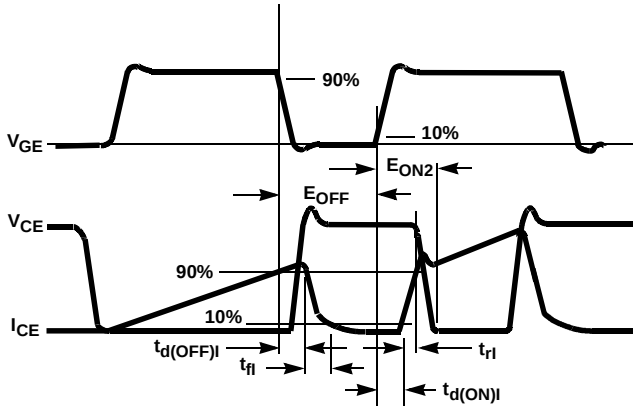


FIGURE 21. SWITCHING TEST WAVEFORMS

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Handling Precautions for IGBTs

Insulated Gate Bipolar Transistors are susceptible to gate-insulation damage by the electrostatic discharge of energy through the devices. When handling these devices, care should be exercised to assure that the static charge built in the handler's body capacitance is not discharged through the device. With proper handling and application procedures, however, IGBTs are currently being extensively used in production by numerous equipment manufacturers in military, industrial and consumer applications, with virtually no damage problems due to electrostatic discharge. IGBTs can be handled safely if the following basic precautions are taken:

1. Prior to assembly into a circuit, all leads should be kept shorted together either by the use of metal shorting springs or by the insertion into conductive material such as "ECCOSORBD LD26" or equivalent.
2. When devices are removed by hand from their carriers, the hand being used should be grounded by any suitable means - for example, with a metallic wristband.
3. Tips of soldering irons should be grounded.
4. Devices should never be inserted into or removed from circuits with power on.
5. **Gate Voltage Rating** - Never exceed the gate-voltage rating of V_{GEM}. Exceeding the rated V_{GE} can result in permanent damage to the oxide layer in the gate region.
6. **Gate Termination** - The gates of these devices are essentially capacitors. Circuits that leave the gate open-circuited or floating should be avoided. These conditions can result in turn-on of the device due to voltage buildup on the input capacitor due to leakage currents or pickup.
7. **Gate Protection** - These devices do not have an internal monolithic Zener diode from gate to emitter. If gate protection is required an external Zener is recommended.

Operating Frequency Information

Operating frequency information for a typical device (Figure 3) is presented as a guide for estimating device performance for a specific application. Other typical frequency vs collector current (I_{CE}) plots are possible using the information shown for a typical unit in Figures 5, 6, 7, 8, 9 and 11. The operating frequency plot (Figure 3) of a typical device shows f_{MAX1} or f_{MAX2}; whichever is smaller at each point. The information is based on measurements of a typical device and is bounded by the maximum rated junction temperature.

f_{MAX1} is defined by $f_{MAX1} = 0.05/(t_{d(OFF)I} + t_{d(ON)I})$. Deadtime (the denominator) has been arbitrarily held to 10% of the on-state time for a 50% duty factor. Other definitions are possible. $t_{d(OFF)I}$ and $t_{d(ON)I}$ are defined in Figure 21. Device turn-off delay can establish an additional frequency limiting condition for an application other than T_{JM}.

f_{MAX2} is defined by $f_{MAX2} = (P_D - P_C)/(E_{OFF} + E_{ON2})$. The allowable dissipation (P_D) is defined by $P_D = (T_{JM} - T_C)/R_{\theta JC}$. The sum of device switching and conduction losses must not exceed P_D. A 50% duty factor was used (Figure 3) and the conduction losses (P_C) are approximated by $P_C = (V_{CE} \times I_{CE})/2$.

E_{ON2} and E_{OFF} are defined in the switching waveforms shown in Figure 21. E_{ON2} is the integral of the instantaneous power loss ($I_{CE} \times V_{CE}$) during turn-on and E_{OFF} is the integral of the instantaneous power loss ($I_{CE} \times V_{CE}$) during turn-off. All tail losses are included in the calculation for E_{OFF}; i.e., the collector current equals zero ($I_{CE} = 0$).

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